

Efe Yılmaz

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SUMMARY

Backend-focused developer and Digital Game Design graduate with internship experience building REST APIs in C# / ASP.NET Core and developing a Unity mobile game. Strong grounding in layered architecture, Entity Framework Core, PostgreSQL, Redis, JWT authentication, Docker, and automated testing with CI; built a real-time matchmaking service load-tested with 1,000 concurrent players. Seeking a junior backend developer role.

EDUCATION

Istinye University — Bachelor's Degree, Digital Game Design · GPA 3.21 / 4.00

Istanbul · 2022 – 2026

EXPERIENCE

Software Developer Intern · *KPlus Bilişim Teknolojileri A.Ş. / Asteri Game Studio*

Istanbul · Jun 2025 – Oct 2025

- Developed and tested backend features for business and application workflows using C#, ASP.NET Core, REST APIs, and SQL database operations.
- Built **Merge Vacuum**, a mobile casual puzzle game in Unity (C#, URP) for Asteri Game Studio, KPlus's in-house game studio, implementing the full gameplay loop, level-objective systems, and game-state UI flow across a 100+ script codebase.

PROJECTS

MatchForge — Real-Time Matchmaking & Lobby Service — ASP.NET Core, SignalR, Redis, PostgreSQL, nginx, k6, Docker, xUnit · github.com/Efekoo/MatchForge

- Built a real-time matchmaking & lobby service closing the full player loop: JWT auth with refresh-token rotation, skill-based matchmaking with expanding Elo/MMR windows, SignalR lobbies, and hand-rolled Elo rating persisted to PostgreSQL.
- Solved race conditions (duplicate matching, concurrent moves) with Redis distributed locks; implemented disconnect grace periods with reconnect and win-by-forfeit handling.
- Load-tested with k6: **1,000 concurrent players matched with p95 match time of 2.0 s and 0% failed requests** across 2 nginx load-balanced replicas with a SignalR Redis backplane; Testcontainers integration tests (real Postgres/Redis) on GitHub Actions CI.

Game Services API — ASP.NET Core Web API, C#, EF Core, PostgreSQL, SignalR, Docker, xUnit · github.com/Efekoo/Game_Services_API · **live demo**

- Designed a four-layer solution (API / Business / Core / DataAccess) exposing seven domain controllers: authentication, user management, currency, inventory, market economy, friends, and leaderboards.
- Implemented JWT authentication with BCrypt password hashing; used Entity Framework Core against PostgreSQL for all data access, with Swagger documentation for every endpoint.
- Built a SignalR real-time leaderboard broadcasting live standings; containerized with Docker Compose and covered by 24 xUnit tests on a GitHub Actions CI pipeline.

Investor — Algorithmic Trading Bot — Python, ccxt, pandas, SQLite, Optuna, Docker, pytest · github.com/Efekoo/Investor

- Built a modular trading bot: exchange abstraction over ccxt, eight pluggable strategies, and a risk module covering position sizing, stop-loss / trailing stop, and a daily circuit breaker.
- Developed a backtest engine simulating fees, slippage, and partial fills with Optuna optimization; designed a layered live-trading safety model (paper / live modes, kill switch); 19 pytest tests run in GitHub Actions CI.

AeroRecover — Python, LangGraph, Google Gemini, Streamlit, Folium · github.com/Efekoo/AeroRecover

- Multi-agent AI application generating post-flight recovery plans for pilots; six specialized agents orchestrated in a LangGraph workflow with safety filtering, delivered as a Streamlit app.

TECHNICAL SKILLS

- **Backend:** C#, ASP.NET Core, Entity Framework Core, REST APIs, PostgreSQL, Redis, SQL Server, SignalR, JWT Authentication, BCrypt, Swagger / OpenAPI, AutoMapper, FluentValidation
- **Testing & DevOps:** xUnit, Testcontainers, pytest, k6 Load Testing, Docker, Docker Compose, nginx, GitHub